When Diverse Paraphrases Do Not Help: LLM Data Augmentation for Portuguese Aspect-Based Sentiment Analysis

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PTT5 baseline · aspect extraction (F1)
0.832
vs 0.637 for the best ABSAPT 2024 system

PTT5 baseline · sentiment (BAcc)
0.775
vs 0.652 for the best ABSAPT 2024 system

Augmented setups that beat the baseline
0 / 4
2 LLMs × 2 sub-tasks, 5 seeds each

1 Motivation

ABSA extracts each **aspect term** of an opinion and its **polarity**. Portuguese data is scarce, so LLM paraphrases are an appealing way to scale it.

POSITIVE NEGATIVE

O **café da manhã** era ótimo, mas o **quarto** estava sujo.

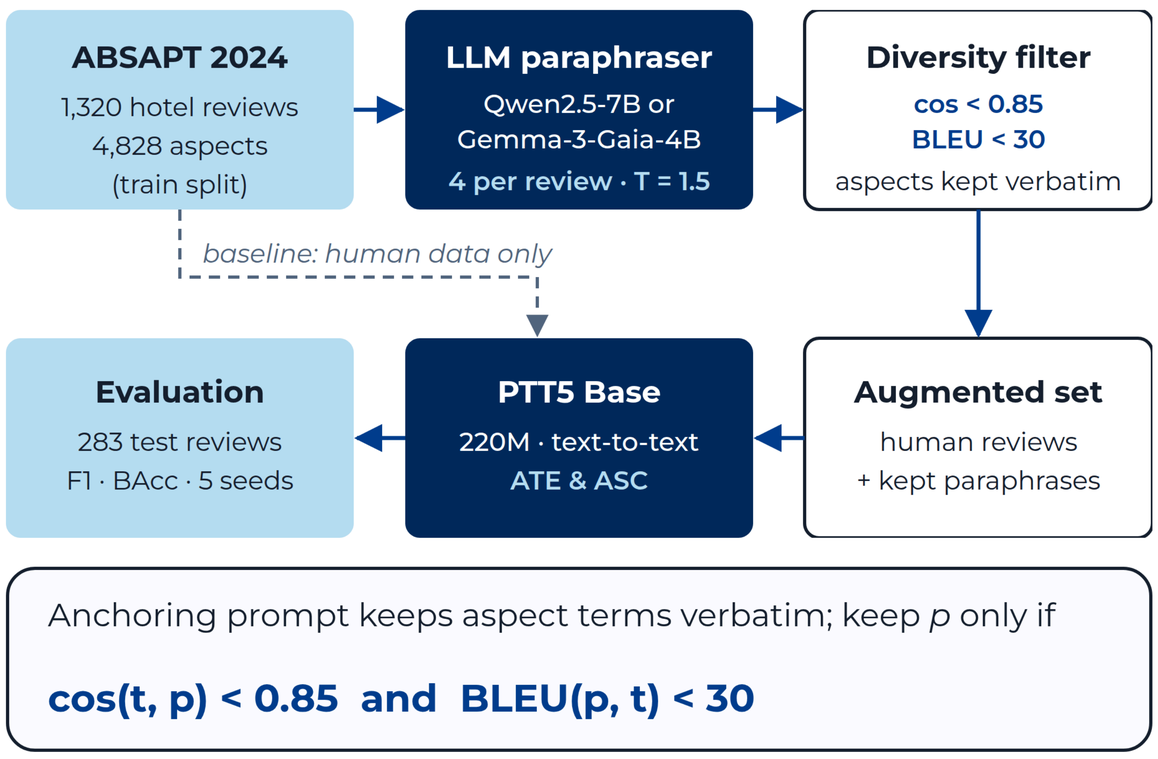
"The breakfast was great, but the room was dirty."

ATE · aspect term extraction | ASC · sentiment

RESEARCH QUESTION

Does in-context LLM paraphrasing yield data that improves a strong model on Portuguese ABSA?

2 Method



Anchoring prompt keeps aspect terms verbatim; keep *p* only if

cos(t, p) < 0.85 and BLEU(p, t) < 30

Paraphraser	Kept / seed	Cosine	BLEU
Qwen2.5-7B	**1193 ± 8**	0.79 ± 0.07	11.2 ± 6.0
Gaia-4B	**1977 ± 30**	0.79 ± 0.07	9.3 ± 5.4

Table 1 — Kept of 5,280 candidates per seed (23% / 37%) and their similarity to the source.

3 Results

Augmentation **never helps**: extraction drops for both LLMs and sentiment stays within seed noise. The paired delta is negative in 4 of 5 seeds for every configuration.

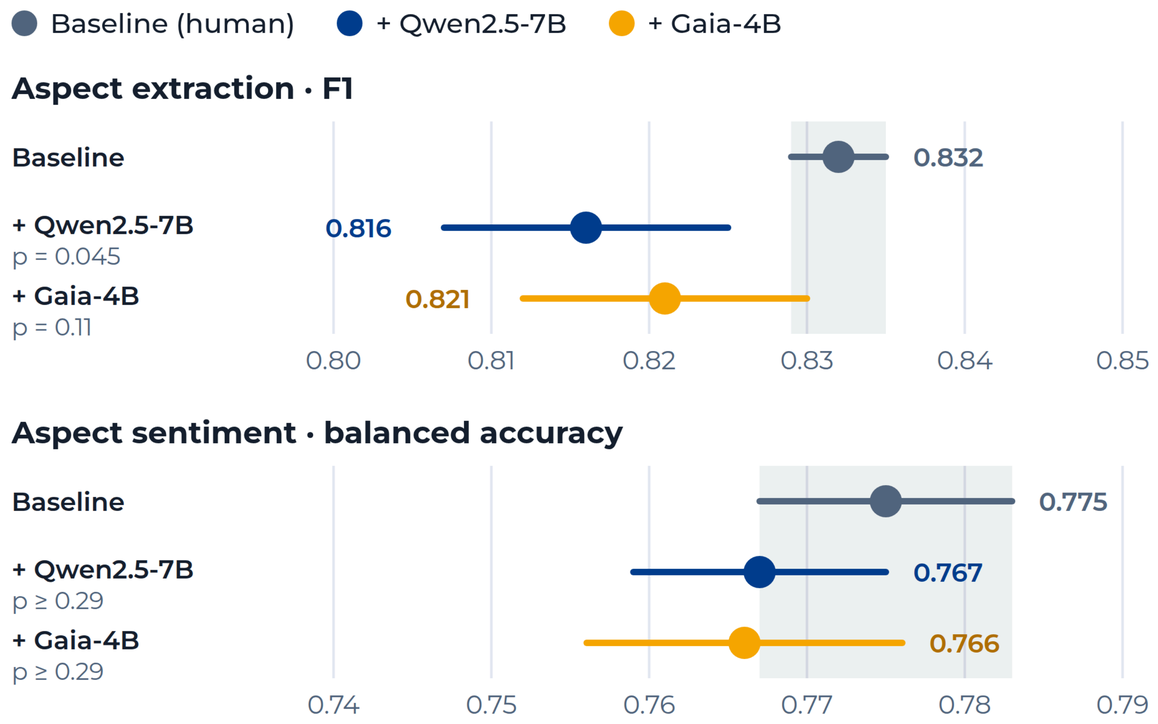


Figure 1 — Mean ± sd over five seeds; shaded band = baseline ± sd. Axes zoomed to a 0.05 range. p: paired t-test over seeds vs baseline.

4 Why no gain?

Training corpus	N	Cosine	BLEU
Human (original)	1317	**0.752**	**5.81**
Qwen2.5-7B	1193	0.725 ± 0.001	4.43 ± 0.08
Gaia-4B	1977	0.730 ± 0.001	4.30 ± 0.03

Table 2 — Proximity of each training corpus to the test set: semantic (max cosine) and surface (BLEU to nearest test neighbours).

Proximity lost vs human data

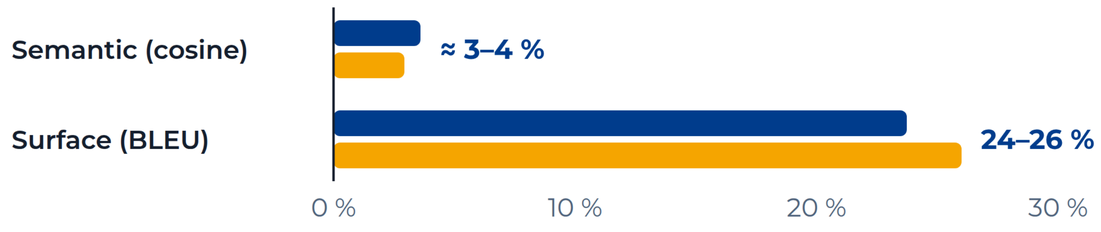


Figure 2 — Relative drop vs the human corpus in Table 2 (blue: Qwen2.5-7B, orange: Gaia-4B). **The injected diversity is mostly lexical.**

TAKE-HOME MESSAGE

Diverse paraphrases do not help: what matters is distance to the evaluation distribution, not distance to the seed.

1. A simple PTT5 baseline beats the best ABSAPT 2024 systems on both sub-tasks.
2. Diverse, aspect-anchored paraphrases do not improve over it, for either LLM.
3. The calibration these heuristics need undercuts the appeal of automated LLM data scaling.

Code & data: **github.com/AndrewwBC/INLG-2026** Contact: **abdcampos@inf.ufpel.edu.br** Funded in part by CAPES (Finance Code 001)